Dr Nadia Mofidi

PhD MSc BSc CEng FIMechE MIFireE
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Current Positions

Technical Director in Fire Safety – Research and Innovation – Sep 2023 - present
Hydrock now Stantec
Visiting Lecturer in Fire Safety – Starting from Jan 2025
Birmingham City University
Guest Lecturer in Fire Safety – Starting from Jan 2025
Cal Poly (California Polytechnic State University)
Visiting Researcher in Civil and Environmental Engineering– Aug 2023 - present
University College London
Member at Steering and Modelling Groups for UK Climate Resilience Roadmap
UK Grean Building Council (UKGBC) – Wildfire studies – Nov 2023 – present
Member of Senior Leadership Team in Fire Division – Nov 2023 – present
Hydrock now Stantec
Member of Technical Development Program for Engineers – Nov 2023 – present
Hydrock now Stantec

Academic Education

PhD in Civil and Environmental Engineering – Fire Science – Sep 2021 – Aug 2024
University College London, UK
Thesis: Air Entrainment Structure in a Turbulent Diffusion Flame Under Cross Wind: A Three-Dimensional Perspective

PhD Candidate at Fire Protection Engineering - Jan 2019 – Aug 2021
Worcester Polytechnic Institute, Worcester, MA, USA

MSc in Fire Protection Engineering – Jan 2017 - Dec 2018
Worcester Polytechnic Institute, Worcester, MA, USA
Research study: Soot Characterization of Wildland Fuels

BSc in Mechanical Engineering – Aug 2004 - Dec 2008
Shahid Rajaee University, Tehran, Iran
Thesis: Developing of a Simulink model for a small-scale jet engine

Publications

- **Flow dynamics of turbulent diffusion flame under cross wind**
  Presented in Proceedings of combustion Institute 2022
- **Soot Characterization of Burning Wildland Porous Fuel Bed**
  11th national meeting of Combustion Institute, CA, USA, Mar 2019
  https://wssci.us/meetings/ncm2019/papers/
- **A study of cavity effect on flame spread over kerosene-soaked sand bed**
  Eastern States Section of the Combustion Institute, PA, USA, Mar 2018
  http://sites.psu.edu/essci2018/downloads/
- **Simulation of Water Mist in ICUs** International Water Mist Conference, Amsterdam, Netherlands, Oct 2015
  http://iwma.net/fileadmin/user_upload/IWMC_2015/Narfoam_Mofidi_IWMC2015_paper.pdf

- **CFD simulation of water mist and sprinkler system in Resalat tunnel of Tehran**
  International Water Mist Conference, Istanbul, Turkey, Oct 2014
  http://iwma.net/fileadmin/user_upload/IWMC_2014/Narfoam_Mofidi_Nadia_IWMC_2014_paper.pdf.

Work in Progress Publications

- **Theoretical study of flame spread over a non-planar kerosene-soaked sand bed**
- **Macro scale air entrainment mechanism in diffusion flame under cross wind**
- **Micro scale air entrainment mechanism in diffusion flame under cross wind**
- **Dimensionless analysis of large-scale diffusion flame under cross wind: revising the boundary layer theory**
- **A methodology to quantify the air entrainment in diffusion flames under cross wind**
- **Flame under cross wind: what is the state-of-the-art and next steps**

Professional Development and Awards

Professional Memberships:

- Fellow of the Institution of Mechanical Engineers (FIMechE), 2024, UK
- Chartered engineer (CEng) and Member of Institution of Mechanical Engineers (MIMechE), 2021, UK
- Member of Institution of Fire Engineers (IFE), 2021, UK (Chartership under review)

Visibility:

- Active STEM Ambassador in the UK and previously in the USA
- Women in construction Ambassador in London Build expo, UK
- Support line volunteer at British Red Cross, 2022, UK

Awards:

- 3rd Place Award Climate Code Red Hackathon competition at UCL, 2021, UK
- 1st Place Award of WPI Graduate Research Innovation Exchange, April 2018, USA
- Extinction in Chemistry and literature Olympiad, 2003, Iran

Scholarships and Grants:

- WPI Graduate Studies Scholarship, September 2017 and 2020, USA
- NFPA Fire Protection Engineering Graduate Support Grant, May 2017, USA
- Maatman Scholarship, James S. Kemper Foundation Fellowship, April 2017, USA

Other:

- Accepted in Combustion Institute summer school at Princeton University, 2017 to 2019, USA
- Endorsed by the Royal Academy of Engineering as Global Promise in Fire Safety, 2021, UK

Previous Work Experience

15+ years of international experience in design, engineering, consulting and fire safety science research – environmental, building and oil & gas industry.

Hydrock, London/Bristol, UK
Technical Director (South West and Wales) April 2023 – November 2023

Hydrock, London/Bristol, UK
Associate Fire Engineer (South West and January 2023 – March 2023

Foster + Partners, London, UK
Senior Fire Engineer (Associate Partner) April 2022 – December 2022

Foster + Partners, London, UK
Senior Fire Engineer (Associate) February 2020 – March 2022

Oskar Brands (Startup), USA
Researcher June 2019 - January 2020

Engineering Planning and Management (EPM), Framingham, MA, USA
Fire Engineer January 2018 – August 2018

Fire Protection Engineering and Mechanical Engineering, WPI, Worcester, MA, USA
Teaching Assistant January 2017 - May 2020

Combustion and Fire Science Laboratories, WPI, Worcester, MA, USA
Lab Assistant January 2017 - January 2020

Narfoamkar Co., Tehran, Iran

Technical lead and Project Manager May 2014 - December 2016

Fire Engineer May 2010 - May 2014

FPE Software and Application Skills

Software: Python, Matlab, FDS, PyroSim, AutoSPRINK, Pipenet, FLUENT, , Microsoft Office, Microsoft Project, AutoCAD, Solidworks, AutoPLANT, CADWorx, CAESAR II, Revit
Analysis: Desk-based studies, Performance-Based Design, overarching studies of international codes, fire risk assessment, explosion protection, experimental data analysis
Fire protection systems: Gaseous, water-based, Foam-based, dry chemical, twin agent, customized deluge packages, storage tanks, fire pump package

Showcase projects:

1. Battersea Power Station, London
2. Kuwait International Airport, Kuwait
3. Sarvestan & Saadat Abad Oil field Project, Iran
4. Bandar Abbas Gas Condensate Refinery
5. Iran Mall, Iran